STUDYLAB HUB · AP & COLLEGE BIOLOGY · UNIT 3 (CHAPTERS 8–10)

AP Biology Unit 3: Cellular Energetics

Bioenergetics · Enzyme Kinetics · Photosynthesis · Cellular Respiration · Practice FRQs & Cram Sheet

AP Biology Unit 3: Cellular Energetics

An Original Study Guide for Understanding Energy, Enzymes, Photosynthesis, and Respiration

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How to Use This Guide

Unit 3 becomes easier when you stop treating it as a list of pathways. Every topic answers the same three questions:

1. What energy transformation is occurring?
2. What molecule or structure makes the transformation possible?
3. How does the cell regulate the process and respond when conditions change?

Read the concept explanations first. Then redraw the diagrams from memory, answer the practice prompts without looking back, and use the cram sheet only during final review.

Part I — The Big Picture

1. Energy in Living Systems

Cells are constantly rearranging matter. They build molecules, move ions, transport cargo, repair membranes, and copy genetic material. These jobs require energy, but a cell cannot simply use heat from its surroundings to power a precise molecular machine. Instead, cells transfer energy through controlled reactions and temporary energy carriers.

A useful model is a series of linked transfers:

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energy source → intermediate carrier → molecular work → heat released
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Food molecules and light are energy sources. ATP and reduced electron carriers are intermediate carriers. Muscle contraction, active transport, synthesis, and movement are examples of work. Heat is released because no energy transfer is perfectly efficient.

Free energy and reaction direction

A reaction with a negative change in free energy can proceed spontaneously under the stated conditions, although it may still need an activation-energy input to begin. A reaction with a positive change in free energy requires an energy input. Cells often link the two types so that energy released by one reaction drives another.

Do not confuse “spontaneous” with “fast.” A reaction can be energetically favorable but extremely slow if its activation barrier is high. Enzymes solve the speed problem; they do not change the overall free-energy difference or the final equilibrium.

Energy is not recycled in the same way as matter

Atoms such as carbon, oxygen, and phosphorus can move through cycles and be reused. Energy enters many ecosystems as sunlight or chemical energy and eventually leaves as heat. Organisms maintain order locally by increasing disorder in their surroundings through energy-releasing processes.

Diagram description: an energy-transfer map

Draw three boxes from left to right. Label the first “source,” the second “carrier,” and the third “cellular work.” Add arrows between them. Under the arrows, write examples: light or food; ATP or NADH; transport, synthesis, or movement. Draw a downward arrow from every box labeled “heat released.” The point is that energy moves through the system, while carriers are repeatedly charged and discharged.

2. ATP as a Transferable Energy Currency

ATP consists of adenine, a ribose sugar, and a chain of three phosphate groups. The cell does not use ATP because the terminal phosphate bond is a magical “high-energy bond.” Energy change depends on the entire reaction, including the stability of products and the surrounding water. Hydrolysis is useful because the products are more stable under cellular conditions and can participate in coupled reactions.

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ATP + H₂O → ADP + phosphate + usable free energy
ADP + phosphate + energy input → ATP
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ATP hydrolysis can be coupled to an energy-requiring reaction. One common mechanism is phosphorylation: a phosphate group is transferred to a substrate or protein. This can alter shape, charge, or reactivity and make a difficult step possible.

Coupling example

Suppose reaction A releases energy and reaction B needs energy. The cell uses an intermediate, often ATP, so that the combined process has a favorable energy change. The reactions are not literally fused into one collision; an enzyme or shared intermediate provides a route connecting them.

Check your understanding

A reaction has a positive free-energy change by itself. Explain why adding ATP may allow it to proceed in a cell. A strong answer mentions coupling, ATP hydrolysis or phosphate transfer, and a combined reaction with a favorable energy change.

Part II — Enzymes and Regulation

3. Enzyme Structure and Specificity

Most enzymes are proteins whose amino-acid sequences fold into particular three-dimensional shapes. A small region called the active site binds one or more substrates. The active site is not simply a rigid hole. Flexible side chains can shift position as the substrate binds, creating an induced fit that places chemical groups in productive orientations.

Specificity comes from several features working together:

- shape compatibility;
- attractions among charged or polar groups;
- hydrophobic interactions;
- hydrogen bonding;
- precise positioning of catalytic side chains.

A substrate can fit an active site and still fail to react if the necessary chemical groups are not aligned or if the transition state is not stabilized.

4. How Enzymes Lower Activation Energy

Every reaction must pass through a high-energy transition state. The activation energy is the energy needed to reach that state. Enzymes lower the barrier by providing a different reaction pathway.

An enzyme may:

- hold reactants close together;
- orient them correctly;
- strain a bond so it is easier to break;
- create a favorable local pH or charge environment;
- stabilize the transition state;
- briefly form a temporary covalent interaction.

The enzyme is regenerated after the reaction. It is not consumed as a reactant, although it can be damaged or inhibited.

Energy diagram description

Draw two curves beginning at the same reactant energy and ending at the same product energy. Make the uncatalyzed curve taller and label its peak “larger activation energy.” Make the catalyzed curve shorter and label its peak “smaller activation energy.” Keep the beginning and ending levels unchanged. This shows that an enzyme changes activation energy, not the reaction’s net free-energy change.

5. Environmental Effects on Enzyme Activity

Temperature

As temperature rises, molecules generally move faster and collide more often, so activity can increase. Beyond an enzyme’s useful range, weak interactions holding the protein’s shape can be disrupted. The active site changes shape, and activity drops sharply. Cooling usually slows movement without permanently unfolding the protein; reheating may restore activity if the enzyme was not damaged.

pH

pH changes the protonation state and charge of amino-acid side chains. Because active-site chemistry depends on charge and shape, an enzyme often has a narrow pH range with highest activity. Extreme pH can disrupt ionic attractions and alter folding.

Substrate concentration

When enzyme concentration is fixed, increasing substrate concentration usually increases reaction rate at first. Eventually most active sites are occupied. The rate approaches a maximum because adding more substrate cannot make the existing enzymes process molecules infinitely fast.

Enzyme concentration

If substrate is abundant, increasing enzyme concentration supplies more active sites and can raise the maximum rate. If substrate becomes limiting, extra enzyme has little effect because there are not enough substrate molecules to occupy it.

Salinity and ionic conditions

Changing salt concentration changes electrostatic interactions. An enzyme from a salt-adapted organism may work best at high salinity, while another enzyme may lose its structure there. Never assume that every enzyme has the same optimum.

Interpreting an activity graph

Always identify the independent variable, dependent variable, and controlled conditions. A curve with a peak suggests an optimum. A plateau suggests a limiting factor. A sudden irreversible decline may indicate denaturation, but the data must support that interpretation.

6. Inhibition and Allosteric Regulation

A competitive inhibitor resembles the substrate or occupies the active site. Increasing substrate can reduce its effect because the two molecules compete for the same location.

A noncompetitive or allosteric inhibitor binds elsewhere and changes enzyme shape. Adding more substrate may not restore the original rate because the active site itself has been altered. In real systems, inhibition categories can be more nuanced than the simplified classroom model; focus on binding location and whether substrate competition explains the data.

Allosteric regulators bind at a site separate from the active site. An activator stabilizes a more active form, while an inhibitor stabilizes a less active form. Feedback inhibition is a pathway-level example: the final product reduces activity of an early enzyme, preventing unnecessary production.

Diagram description: feedback inhibition

Draw a row of four enzyme-controlled steps: A → B → C → D. Draw an arrow from D backward to the first enzyme. Label the backward arrow “final product inhibits early step.” Add a note: “prevents waste when D is abundant.”

7. Experimental Design with Enzymes

A strong enzyme investigation includes:

- one independent variable, such as temperature or pH;
- a measurable dependent variable, such as product formed per minute;

- a control or baseline condition;
- repeated trials;
- constant enzyme and substrate amounts unless one is being tested;
- a clear operational definition of “activity.”

If testing temperature, pre-equilibrate both enzyme and substrate, or explain why the setup does not. Otherwise the samples may not begin at comparable temperatures. If using foam height as a proxy for gas production, state that foam height is an indirect measure and may be affected by bubble stability.

Practice mini-investigation

Students test an enzyme at pH 4, 6, 8, and 10. The average product rates are 3, 11, 18, and 5 units per minute.

Questions:

1. Which pH appears optimal?
2. What claim is supported by the data?
3. Name one variable that should remain constant.
4. Why should the students repeat each condition?

Answer: pH 8 appears optimal. The enzyme’s measured activity is highest at pH 8 under these conditions. Enzyme concentration, substrate concentration, temperature, reaction time, or volume could be controlled. Repetition helps estimate variation and makes the comparison more reliable.

Part III — Photosynthesis

8. Photosynthesis as Energy Storage

Photosynthesis converts light energy into chemical energy stored in carbohydrates. In plants and algae, chloroplasts separate major stages spatially. The light-dependent reactions occur in thylakoid membranes. The Calvin cycle occurs in the surrounding fluid, the stroma.

A simplified overall equation is:

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carbon dioxide + water + light energy → carbohydrate + oxygen
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This summary hides important details. The oxygen released comes from splitting water, not from carbon dioxide. Carbon dioxide supplies carbon that becomes part of carbohydrate. Light energy is first captured by pigments and transferred through electron carriers.

9. Pigments and Absorption

Pigments absorb particular wavelengths. Chlorophyll a is central to the reaction-center chemistry, while accessory pigments absorb additional wavelengths and transfer excitation energy. A leaf appears green because green wavelengths are reflected or transmitted more than red and blue wavelengths.

An absorption spectrum shows which wavelengths a pigment absorbs. An action spectrum shows photosynthetic output across wavelengths. The two patterns often resemble each other but need not match perfectly because several pigments contribute.

10. Light-Dependent Reactions

Photons excite electrons in pigment molecules. The reaction centers pass energized electrons to acceptors, beginning an electron-transport sequence. As electrons move through carriers, released energy helps pump hydrogen ions across the thylakoid membrane. This creates an electrochemical gradient.

Hydrogen ions flow back through ATP synthase. The enzyme uses the gradient to produce ATP from ADP and phosphate. Electrons ultimately reduce $NADP^+$ to NADPH. ATP and NADPH then provide energy and reducing power for carbon fixation reactions.

Linear electron flow

In linear flow, electrons begin in water and end in NADPH. Water splitting replaces electrons lost from the first photosystem and releases oxygen. Both photosystems participate. The chain produces ATP and NADPH.

Cyclic electron flow

Cyclic flow sends excited electrons through carriers and back to the same photosystem. It increases ATP production without producing NADPH or oxygen. This can help balance the ATP demand of the Calvin cycle.

Diagram description: a thylakoid “battery”

Draw a membrane as two horizontal lines. Label the space above “stroma” and below “thylakoid interior.” Place two light-capturing complexes in the membrane. Show water entering the first complex and oxygen leaving.

Draw electrons moving through carriers toward the second complex and then to $NADP^+$. Show hydrogen ions accumulating inside. Draw ATP synthase as a channel with H^+ arrows flowing out toward the stroma and ATP forming on the stroma side.

11. The Calvin Cycle

The Calvin cycle uses ATP and NADPH to incorporate carbon dioxide into organic molecules. It has three conceptual phases:

1. Carbon fixation: carbon dioxide is attached to a five-carbon acceptor, producing unstable intermediates that split into three-carbon molecules.
2. Reduction: ATP and NADPH raise the energy and electron content of the three-carbon molecules.
3. Regeneration: some three-carbon molecules leave as carbohydrate building material, while the rest use ATP to regenerate the five-carbon acceptor.

The cycle does not directly require photons, but it depends on products of the light-dependent reactions. If light reactions stop, ATP and NADPH supplies decline and carbon fixation cannot continue for long.

For every three carbon dioxide molecules entering, one net three-carbon sugar molecule can leave after the cycle’s accounting steps. More carbon dioxide entries are needed to build a six-carbon sugar.

Common misconception

The Calvin cycle is not “dark” because it only runs at night. It is called light-independent because photons are not directly absorbed by its enzymes. In a living leaf, it usually depends on daytime products and regulation linked to light conditions.

12. Photosynthesis Data Analysis

A rate of photosynthesis can be estimated from oxygen production, carbon dioxide uptake, or bubble formation. In aquatic plant experiments, bubbles are often an indirect proxy and should be interpreted carefully.

If light intensity increases and the rate rises before leveling off, light was limiting at first and another factor became limiting later. If adding carbon dioxide increases the rate, carbon availability was limiting. If no treatment changes the rate, check whether the measurement is sensitive enough and whether another factor is already limiting.

Part IV — Cellular Respiration

13. Respiration as Controlled Oxidation

Cellular respiration transfers energy from reduced carbon compounds to ATP. Glucose is not burned in one step. Instead, electrons are removed through a sequence of reactions, and their energy is temporarily carried by NADH and $FADH_2$.

A simplified summary is:

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glucose + oxygen → carbon dioxide + water + ATP + heat
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The glucose carbon atoms become carbon dioxide. Oxygen accepts electrons near the end of the electron-transport chain and combines with hydrogen ions to form water.

14. Glycolysis

Glycolysis occurs in the cytosol and does not require oxygen directly. One six-carbon glucose is rearranged and split into two three-carbon molecules. The pathway has an investment phase and a payoff phase.

For each glucose, the net outputs are:

- 2 pyruvate;
- 2 NADH;
- 2 ATP net.

Four ATP are produced in the pathway, but two ATP are used earlier, so the net gain is two. ATP is made by substrate-level phosphorylation, in which an enzyme transfers phosphate to ADP from a phosphorylated intermediate.

15. Pyruvate Oxidation

In eukaryotic cells, pyruvate enters the mitochondrial matrix. Each pyruvate loses one carbon as carbon dioxide.

The remaining two-carbon fragment joins coenzyme A, and NAD^+ is reduced to NADH.

Per glucose, because two pyruvate are formed, this step produces:

- 2 acetyl-CoA;
- 2 carbon dioxide;
- 2 NADH.

This is a transition step: the carbon skeleton is prepared for the citric acid cycle, and more electrons are loaded onto carriers.

16. Citric Acid Cycle

Acetyl-CoA contributes a two-carbon group to a cycle of reactions in the mitochondrial matrix. The starting acceptor is regenerated, allowing the cycle to repeat. Per glucose, the cycle produces:

- 4 carbon dioxide;
- 6 NADH;
- 2 $FADH_2$;
- 2 ATP or an equivalent nucleotide triphosphate.

The cycle's most important role is not the small ATP yield. It loads many electrons onto NADH and $FADH_2$, which deliver them to the electron-transport chain.

Diagram description: carbon accounting

Draw glucose on the left and two pyruvate below it. From pyruvate, draw two carbon dioxide arrows and two acetyl-CoA arrows. Draw acetyl-CoA entering a circular pathway. Around the circle, place labels for NADH, $FADH_2$, ATP, and carbon dioxide. Use "per glucose" beside the totals so the reader does not confuse them with totals per turn.

17. Oxidative Phosphorylation

Oxidative phosphorylation includes the electron-transport chain and chemiosmosis. In eukaryotes, the chain is embedded in the inner mitochondrial membrane.

NADH and $FADH_2$ donate electrons to membrane proteins. As electrons move through the chain, energy is used to pump hydrogen ions from the matrix into the intermembrane space. This produces both a concentration gradient and an electrical gradient.

Oxygen is the terminal electron acceptor. Without oxygen, electrons back up, proton pumping slows, and oxidative ATP production stops. Oxygen is not used to split glucose at the beginning; it is needed at the end of the chain.

18. Chemiosmosis and ATP Synthase

Chemiosmosis is the use of an ion gradient to drive work. Hydrogen ions flow down their electrochemical gradient through ATP synthase. The movement causes conformational changes in the enzyme, allowing ADP and phosphate to form ATP.

The inner mitochondrial membrane must be relatively impermeable to hydrogen ions for the gradient to remain useful. If the membrane becomes leaky, the chain may still transfer electrons, but less of the released energy is captured as ATP.

Uncoupling

An uncoupling molecule provides an alternate path for hydrogen ions across the membrane. The gradient is reduced, ATP production becomes less efficient, and more energy is released as heat. This illustrates that electron transport and ATP synthesis are linked but not identical processes.

19. ATP Yield and Efficiency

Exact ATP totals can vary among cells because electrons from cytosolic NADH may enter mitochondria through different shuttle systems, and some energy supports transport or other costs. Avoid treating one number as universal. The reliable comparison is that oxidative phosphorylation generally produces far more ATP per glucose than glycolysis alone, while substrate-level phosphorylation produces ATP directly in specific reaction steps.

Aerobic versus anaerobic conditions

With oxygen available, pyruvate can be processed further and electrons can be transferred to oxygen through the chain. Without oxygen, cells need another way to regenerate NAD^+ so glycolysis can continue.

Part V — Fermentation and Comparisons

20. Fermentation

Fermentation does not create a large additional ATP yield. Its central purpose is to recycle NADH back into NAD^+ by transferring electrons to an organic molecule.

In lactic acid fermentation, pyruvate receives electrons and becomes lactate. In alcoholic fermentation, pyruvate is converted through an intermediate to ethanol, releasing carbon dioxide along the way. The exact products differ, but the logic is the same: glycolysis can keep running because NAD^+ is restored.

The net ATP yield from fermentation-supported glycolysis is two ATP per glucose. Fermentation is useful for short-term or oxygen-limited metabolism, but it extracts much less energy from glucose than aerobic respiration.

21. Comparison Table

Feature	Aerobic respiration	Fermentation
Oxygen required directly?	Yes, as terminal acceptor	No
Glycolysis occurs?	Yes	Yes
Main ATP source	Oxidative phosphorylation plus substrate-level phosphorylation	Substrate-level phosphorylation in glycolysis
NAD^+ regeneration	Electron-transport chain	Organic molecule accepts electrons
Carbon products	Mostly carbon dioxide	Lactate, or ethanol plus carbon dioxide
ATP per glucose	Much greater	Net 2
Useful when	Oxygen and functioning mitochondria or chains are available	Oxygen is absent or electron transport is unavailable

22. Photosynthesis and Respiration Together

Photosynthesis stores energy in carbon compounds. Respiration transfers that stored energy into ATP. The products of one process can serve as reactants of the other, but the pathways are not simple reversals.

Photosynthesis uses light to reduce carbon. Respiration oxidizes carbon and transfers electrons to carriers. The two processes occur in different compartments and use different enzyme systems. Both rely on membranes, electron carriers, ion gradients, and ATP synthase-like molecular machines.

Part VI — Free-Response-Style Practice

These items are original practice prompts written for this guide. They are not released AP questions.

Practice 1: Enzyme temperature investigation

A student measures the rate of product formation for an enzyme at 10°C, 25°C, 40°C, 55°C, and 70°C. The rate rises from 10°C to 40°C, then drops sharply at 55°C and is nearly zero at 70°C.

A. Describe the relationship between temperature and enzyme rate.

B. Explain the increase between 10°C and 40°C.

C. Explain one possible cause of the decline above 40°C.

D. Design a control condition that would help determine whether the loss at 70°C is reversible.

Key

A. Rate increases to an apparent optimum near 40°C and then decreases sharply. B. Molecules move faster and collisions become more frequent, increasing successful enzyme-substrate encounters. C. High temperature can disrupt interactions that maintain protein folding, changing the active site. D. Heat-treated enzyme could be cooled and tested again under a standard temperature; recovery would suggest reversible loss, while continued low activity would support permanent denaturation or damage.

Practice 2: Inhibitor reasoning

Two groups measure the same enzyme. Group 1 adds increasing substrate concentrations. Group 2 adds increasing concentrations of a molecule that binds at an allosteric site. Group 2 reaches a lower maximum rate.

A. Identify the likely inhibition pattern.

B. Predict what happens if substrate is added to Group 2.

C. Explain why the prediction follows from the inhibitor’s binding location.

Key

A. Allosteric or noncompetitive-like inhibition. B. Adding substrate may not restore the original maximum rate. C. The inhibitor changes enzyme conformation from a site separate from the active site, so occupying more active sites with substrate does not necessarily remove the altered shape.

Practice 3: Photosynthetic limiting factor

A plant is tested at three light intensities. Oxygen production rises from low to medium light but is nearly identical at medium and high light. A second trial at high light adds carbon dioxide, and oxygen production increases.

A. Identify the limiting factor at low light.

B. Identify the likely limiting factor at high light before carbon dioxide is added.

C. Explain why oxygen production increased after carbon dioxide was added.

D. Predict what would happen if temperature were raised far above the plant's normal range.

Key

A. Light availability. B. Carbon dioxide availability or another factor downstream of light capture. C. Carbon dioxide was limiting carbon fixation, so increasing it allowed more use of ATP and NADPH and increased overall photosynthetic throughput. D. The rate could decline if high temperature disrupts enzyme structure, membrane function, or water balance.

Practice 4: Electron-flow mutation

A mutation prevents a photosynthetic electron carrier from transferring electrons between two complexes.

A. Predict the effect on the hydrogen-ion gradient.

B. Predict the effect on ATP production by the photosynthetic ATP synthase.

C. Predict the effect on NADPH production.

D. Explain why oxygen may still be produced briefly after the mutation but decline later.

Key

A. The gradient should become smaller because electron flow supplies less energy for ion pumping. B. ATP production should decrease because ATP synthase has less gradient energy. C. NADPH production should decrease because fewer electrons reach the final acceptor. D. Water splitting may continue temporarily while the system contains usable electron acceptors, but electron carriers eventually become over-reduced and flow slows or stops.

Practice 5: Respiration without oxygen

A muscle cell suddenly experiences a severe oxygen shortage.

A. Explain why oxidative phosphorylation decreases.

B. Explain how fermentation allows glycolysis to continue.

C. Compare the net ATP yield per glucose with normal aerobic metabolism.

D. Explain why fermentation cannot replace aerobic respiration for long-term high energy demand.

Key

A. Oxygen cannot accept electrons at the end of the chain, so electron carriers remain reduced and proton pumping declines. B. Fermentation transfers electrons from NADH to pyruvate or a derivative, regenerating NAD^+ for glycolysis. C. Fermentation-supported glycolysis yields a net two ATP per glucose, much less than

complete aerobic processing. D. It extracts less energy per glucose and can cause accumulation of end products, so it is less sustainable for high demand.

Practice 6: Membrane leak

A chemical makes the inner mitochondrial membrane permeable to hydrogen ions without passing through ATP synthase.

A. Predict the proton gradient.

B. Predict oxygen consumption if the electron-transport chain remains functional.

C. Predict ATP production.

D. Identify where some of the captured energy may go.

Key

A. The gradient decreases because ions leak back across the membrane. B. Oxygen consumption may continue or even increase as the chain attempts to restore the gradient. C. ATP production decreases because fewer ions pass through ATP synthase. D. More energy is released as heat.

Part VII — Vocabulary

Activation energy — energy barrier that reactants must overcome.

Active site — region of an enzyme where substrate binding and catalysis occur.

ADP — adenosine diphosphate; a lower-phosphate form that can be phosphorylated to ATP.

Allosteric regulation — control caused by binding at a site other than the active site.

ATP — adenosine triphosphate; a transferable energy carrier used in cellular work.

Calvin cycle — carbon-fixation pathway that uses ATP and NADPH to build carbohydrate precursors.

Carbon fixation — incorporation of inorganic carbon dioxide into an organic molecule.

Catalyst — substance that speeds a reaction without being permanently consumed.

Chemiosmosis — use of an ion gradient to drive ATP synthesis or other work.

Chloroplast — organelle where photosynthesis occurs in plants and algae.

Competitive inhibitor — molecule that competes with substrate for the active site.

Cyclic electron flow — electron pathway that increases ATP production without directly producing NADPH or oxygen.

Denaturation — loss of functional protein shape.

Electron carrier — molecule that accepts and transfers high-energy electrons.

Electrochemical gradient — combined concentration and electrical difference across a membrane.

Fermentation — pathway that regenerates NAD^+ by transferring electrons to an organic molecule.

Free energy — energy available to perform work under particular conditions.

Glycolysis — cytosolic pathway that splits glucose into pyruvate and produces ATP and NADH.

Hydrogen-ion gradient — unequal distribution of H^+ across a membrane.

Induced fit — shape adjustment of an enzyme as substrate binds.

NADH — reduced electron carrier produced during oxidation of fuel molecules.

NADPH — reduced electron carrier that supplies reducing power for biosynthesis, including the Calvin cycle.

Oxidative phosphorylation — ATP production powered by electron transport and chemiosmosis.

Photosystem — pigment-protein complex that captures light and initiates electron flow.

Proton-motive force — energy stored in a hydrogen-ion electrochemical gradient.

Pyruvate — three-carbon product of glycolysis.

Redox reaction — reaction involving electron transfer; oxidation loses electrons and reduction gains them.

Substrate-level phosphorylation — direct transfer of phosphate to ADP from a phosphorylated intermediate.

Terminal electron acceptor — final molecule receiving electrons in an electron-transport chain.

Thylakoid — internal chloroplast membrane containing photosynthetic electron-transfer machinery.

Transition state — unstable, high-energy arrangement reached during a reaction.

Part VIII — One-Page Cram Sheet

Energy: favorable does not mean fast. Enzymes lower activation energy, not net free-energy change.

ATP: hydrolysis can drive endergonic work through coupling and phosphorylation.

Enzymes: active-site shape and chemistry determine specificity. Temperature, pH, salinity, substrate concentration, and enzyme concentration can affect rate.

Inhibition: competitive inhibitors occupy the active site; allosteric inhibitors change enzyme shape elsewhere. Feedback inhibition prevents overproduction.

Photosynthesis location: light reactions in thylakoid membrane; Calvin cycle in stroma.

Light reactions: water supplies electrons; oxygen is released; electron transport builds an H^+ gradient; ATP synthase makes ATP; $NADP^+$ becomes NADPH.

Calvin cycle: fixation, reduction, regeneration. Uses CO_2, ATP, and NADPH. Produces a three-carbon sugar output after sufficient turns.

Respiration location: glycolysis in cytosol; pyruvate oxidation and citric acid cycle in mitochondrial matrix; electron transport and ATP synthase at inner mitochondrial membrane.

Glycolysis per glucose: 2 pyruvate, 2 NADH, net 2 ATP.

Pyruvate oxidation per glucose: 2 acetyl-CoA, 2 CO_2, 2 NADH.

Citric acid cycle per glucose: 4 CO_2, 6 NADH, 2 $FADH_2$, 2 ATP-equivalent.

Oxidative phosphorylation: NADH/$FADH_2$ donate electrons; chain pumps H^+; oxygen accepts electrons; H^+ flows through ATP synthase.

Fermentation: regenerates NAD^+ so glycolysis continues; does not add a large ATP yield; net 2 ATP per glucose.

Data analysis: name variables, identify controls, distinguish correlation from causation, explain plateaus as possible limiting factors, and avoid claiming denaturation without evidence.

High-value comparison: photosynthesis stores energy by reducing carbon; respiration releases energy by oxidizing carbon. Both use electron carriers, membranes, ion gradients, and ATP synthase.

Review Notes

1. Exact ATP totals vary with cell type, shuttle systems, transport costs, and accounting conventions. Use relative comparisons unless a question defines a convention.
2. “Noncompetitive inhibition” is often taught as a simplified category. Real inhibitor behavior may be mixed; focus on the evidence about binding and maximum rate.
3. The Calvin cycle is light-independent in direct photon use, but its operation in living cells is linked to light-reaction products and light-regulated enzyme states.
4. Oxygen bubbles and foam are indirect measures. A careful experiment should explain what the proxy measures and what could affect it.
5. AP Biology questions may emphasize experimental reasoning and mechanism rather than memorization of every intermediate.

End of guide.